Project: Worked examples
Title: Load combinations (EN 1990)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Design values of a single action effect (a force, a moment or a load intensity) under every EN 1990 Annex A1 combination for buildings: ULS persistent/transient, SLS characteristic, frequent and quasi-permanent, and accidental. Every variable action is tried as the leading one; the maximum and minimum envelopes and their governing combinations are reported.

WHAT IT CALCULATES	RELATION
ULS 6.10	$\Sigma\ \gamma_G \cdot G + \gamma_Q \cdot K_{FI} \cdot Q_1 + \Sigma\ \gamma_Q \cdot K_{FI} \cdot \psi_0 \cdot Q_i$ (K_{FI} also on $\gamma_{G,sup}$)
ULS 6.10a	$\Sigma\ \gamma_G \cdot G + \Sigma\ \gamma_Q \cdot K_{FI} \cdot \psi_0 \cdot Q_i$; 6.10b: $\Sigma\ \xi \cdot \gamma_G \cdot G + \gamma_Q \cdot K_{FI} \cdot Q_1 + \Sigma\ \gamma_Q \cdot K_{FI} \cdot \psi_0 \cdot Q_i$
Permanent factor	$\gamma_{G,sup}$ when the action adds to the effect sought, $\gamma_{G,inf}$ when it opposes it
Variable actions	included only when they add to the effect sought (otherwise zero)
SLS characteristic 6.14b	$\Sigma\ G + Q_1 + \Sigma\ \psi_0 \cdot Q_i$; frequent 6.15b: $\Sigma\ G + \psi_1 \cdot Q_1 + \Sigma\ \psi_2 \cdot Q_i$; quasi-permanent 6.16b: $\Sigma\ G + \Sigma\ \psi_2 \cdot Q_i$
Accidental 6.11b	$\Sigma\ G + A_d + \psi_1 \cdot Q_1 + \Sigma\ \psi_2 \cdot Q_i$

Basis: Combinations for persistent or transient design situations (6.10, 6.10a, 6.10b)[1](cl. 6.4.3.2); Combinations for accidental design situations (6.11b)[1](cl. 6.4.3.3); Serviceability combinations (6.14b, 6.15b, 6.16b)[1](cl. 6.5.3); Application for buildings: ψ and γ factors[1](Annex A1, Tables A1.1, A1.2(B)); UK values of partial factors[1](NA.A1.2(A)–(C)).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
actions	Actions (one effect, signed)	2 row(s)		drawing: GA-L2-001 rev C	any
unit	Unit of the effect	kN/m		assumption: Line load on the beam.	any
method	ULS expression	6.10a and 6.10b		spec: UK NA Table NA.A1.2(B): the less favourable of 6.10a and 6.10b	any
$\gamma_{G,sup}$	$\gamma_{G,sup}$ (unfavourable permanent)	1.35		spec: UK NA Table NA.A1.2(B)	1 … 2
$\gamma_{G,inf}$	$\gamma_{G,inf}$ (favourable permanent)	1		spec: UK NA Table NA.A1.2(B)	0.8 … 1.2
γ_Q	γ_Q (variable)	1.5		spec: UK NA Table NA.A1.2(B)	1 … 2
ξ	ξ reduction for 6.10b	0.925		spec: UK NA Table NA.A1.2(B)	0.8 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
K_{FI}	K_{FI} consequence-class factor	1		assumption: Consequence class CC2.	0.9 … 1.1

TABLE ROW	VALUES
actions[1]	name = Slab, finishes and services; category = G permanent; F_k = 18
actions[2]	name = Office imposed; category = B office; F_k = 10

3. Assumptions

A1. One scalar action effect; every action's characteristic value is its contribution to that effect, with sign. (Linear superposition (EN 1990 cl. 6.3.1 representative values).)
A2. Buildings only (Annex A1); no bridges, fatigue or seismic combinations. (EN 1990 Annex A1 scope.)
A3. Without a ψ record the EN 1990 Annex A1 recommended values are used, labelled on the sheet; the UK National Annex (NA.A1.2(A)–(C)) values come from a user record. (No National Annex table is shipped.)
A4. Each permanent action takes $\gamma_{G,sup}$ or $\gamma_{G,inf}$ by whether it adds to or opposes the effect sought; for EQU single-source checks, enter a source's permanent actions as one row. (EN 1990 Table A1.2(B) note on favourable and unfavourable permanent actions.)
A5. unit (Unit of the effect) is taken as kN/m. (Line load on the beam.)
A6. K_{FI} (K_{FI} consequence-class factor) is taken as 1. (Consequence class CC2.)
A7. ψ factors: EN 1990 Annex A1 recommended values (Table A1.1). (No ψ record entered.)

4. Working

[8] $C1 = \gamma_{G,sup} \cdot K_{FI} \cdot G1 + \gamma_Q \cdot K_{FI} \cdot \psi_0 \cdot Q2$ [1](cl. 6.4.3.2)
$= 1.35 \times 18 + 1.05 \times 10$
$= 34.8$ kN/m
ULS 6.10a (max)

[9] $C2 = \xi \cdot \gamma_{G,sup} \cdot K_{FI} \cdot G1 + \gamma_Q \cdot K_{FI} \cdot Q2$ [1](cl. 6.4.3.2)
$= 1.2488 \times 18 + 1.5 \times 10$
$= 37.478$ kN/m
ULS 6.10b, Office imposed leading (max)

[10] $C3 = 1 \cdot G1 + 1 \cdot Q2$ [1](cl. 6.5.3)
$= 1 \times 18 + 1 \times 10$
$= 28$ kN/m
SLS characteristic 6.14b, Office imposed leading (max)

[11] $C4 = 1 \cdot G1 + \psi_1 \cdot Q2$ [1](cl. 6.5.3)
$= 1 \times 18 + 0.5 \times 10$
$= 23$ kN/m
SLS frequent 6.15b, Office imposed leading (max)

[12] $C5 = 1 \cdot G1 + \psi_2 \cdot Q2$ [1](cl. 6.5.3)
$= 1 \times 18 + 0.3 \times 10$
$= 21$ kN/m
SLS quasi-permanent 6.16b (max)

[13] $D1 = \gamma_{G,inf} \cdot G1$ [1](cl. 6.4.3.2)
$= 1 \times 18$
$= 18$ kN/m
ULS 6.10a (min)

[14] $D2 = \gamma_{G,inf} \cdot G1$ [1](cl. 6.4.3.2)
$= 1 \times 18$
$= 18$ kN/m
ULS 6.10b, no variable action (min)

[15] D3 = 1·G1 [1](cl. 6.5.3)
= 1×18
= 18 kN/m
SLS characteristic 6.14b, no variable action (min)
[16] D4 = 1·G1 [1](cl. 6.5.3)
= 1×18
= 18 kN/m
SLS frequent 6.15b, no variable action (min)
[17] D5 = 1·G1 [1](cl. 6.5.3)
= 1×18
= 18 kN/m
SLS quasi-permanent 6.16b (min)
[18] $E_{d,max}$ = maximum over the combinations
= ULS 6.10b, Office imposed leading
= 37.478 kN/m
ULS design effect, maximum
[19] $E_{d,min}$ = minimum over the combinations
= ULS 6.10a
= 18 kN/m
ULS design effect, minimum
[20] $E_{k,max}$ = maximum over the combinations
= SLS characteristic 6.14b, Office imposed leading
= 28 kN/m
SLS characteristic effect, maximum
[21] $E_{k,min}$ = minimum over the combinations
= SLS characteristic 6.14b, no variable action
= 18 kN/m
SLS characteristic effect, minimum
[22] $E_{fr,max}$ = maximum over the combinations
= SLS frequent 6.15b, Office imposed leading
= 23 kN/m
SLS frequent effect, maximum
[23] $E_{fr,min}$ = minimum over the combinations
= SLS frequent 6.15b, no variable action
= 18 kN/m
SLS frequent effect, minimum
[24] $E_{qp,max}$ = maximum over the combinations
= SLS quasi-permanent 6.16b
= 21 kN/m
SLS quasi-permanent effect, maximum
[25] $E_{qp,min}$ = minimum over the combinations
= SLS quasi-permanent 6.16b
= 18 kN/m
SLS quasi-permanent effect, minimum

5. Results

SYMBOL	VALUE	DESCRIPTION
$E_{d,max}$	37.5 kN/m	ULS design effect, maximum
$E_{d,min}$	18.0 kN/m	ULS design effect, minimum
$E_{k,max}$	28.0 kN/m	SLS characteristic effect, maximum
$E_{qp,max}$	21.0 kN/m	SLS quasi-permanent effect, maximum

SYMBOL	VALUE	DESCRIPTION
$E_{d,gov}$	"ULS 6.10b, Office imposed leading"	ULS design effect, governing (max)
$E_{k,min}$	18.0 kN/m	SLS characteristic effect, minimum
$E_{k,gov}$	"SLS characteristic 6.14b, Office imposed leading"	SLS characteristic effect, governing (max)
$E_{fr,max}$	23.0 kN/m	SLS frequent effect, maximum
$E_{fr,min}$	18.0 kN/m	SLS frequent effect, minimum
$E_{fr,gov}$	"SLS frequent 6.15b, Office imposed leading"	SLS frequent effect, governing (max)
$E_{qp,min}$	18.0 kN/m	SLS quasi-permanent effect, minimum
$E_{qp,gov}$	"SLS quasi-permanent 6.16b"	SLS quasi-permanent effect, governing (max)

6. Checks (governing first)

None.

7. Applicability

WARNING records: tabulated values not taken from a reference record are $\gamma_{G,sup}$, $\gamma_{G,inf}$, γ_Q, ξ (EN 1990 partial and ψ factors). Enter each as one so its source, edition and licence travel with it. [record]

8. Verification

`Template struct.load-combinations@1.1.0 Content hash 28f3ed111e112df1 Engine calc-engine/1.0.0 Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. British Standards Institution. Eurocode: Basis of Structural Design. BS EN 1990. BSI.